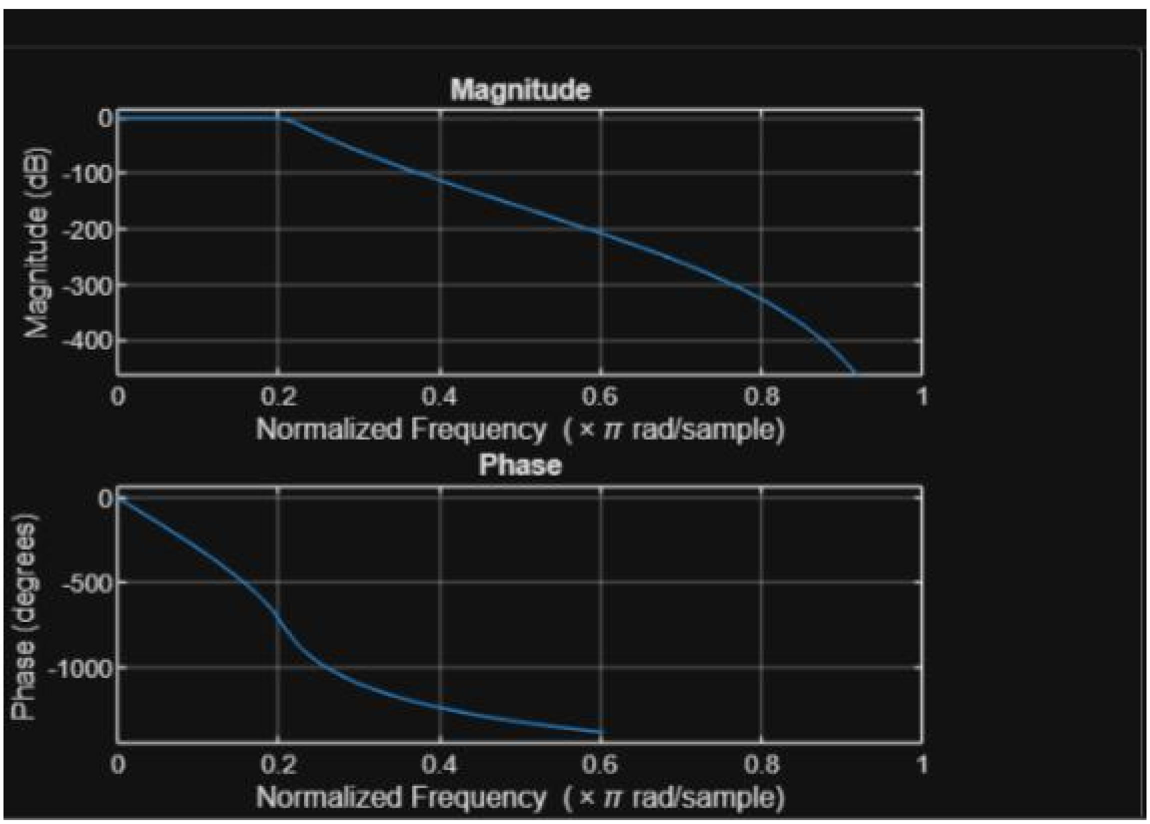
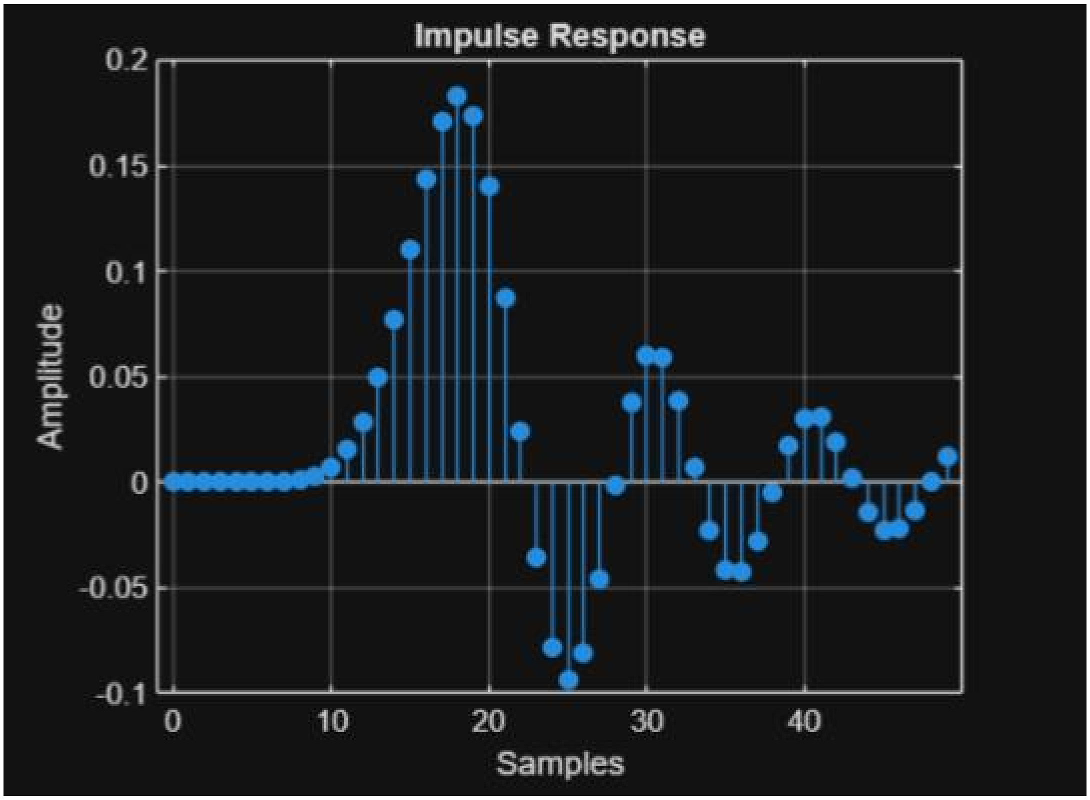
EXP 1.Butterworth IIR Filters

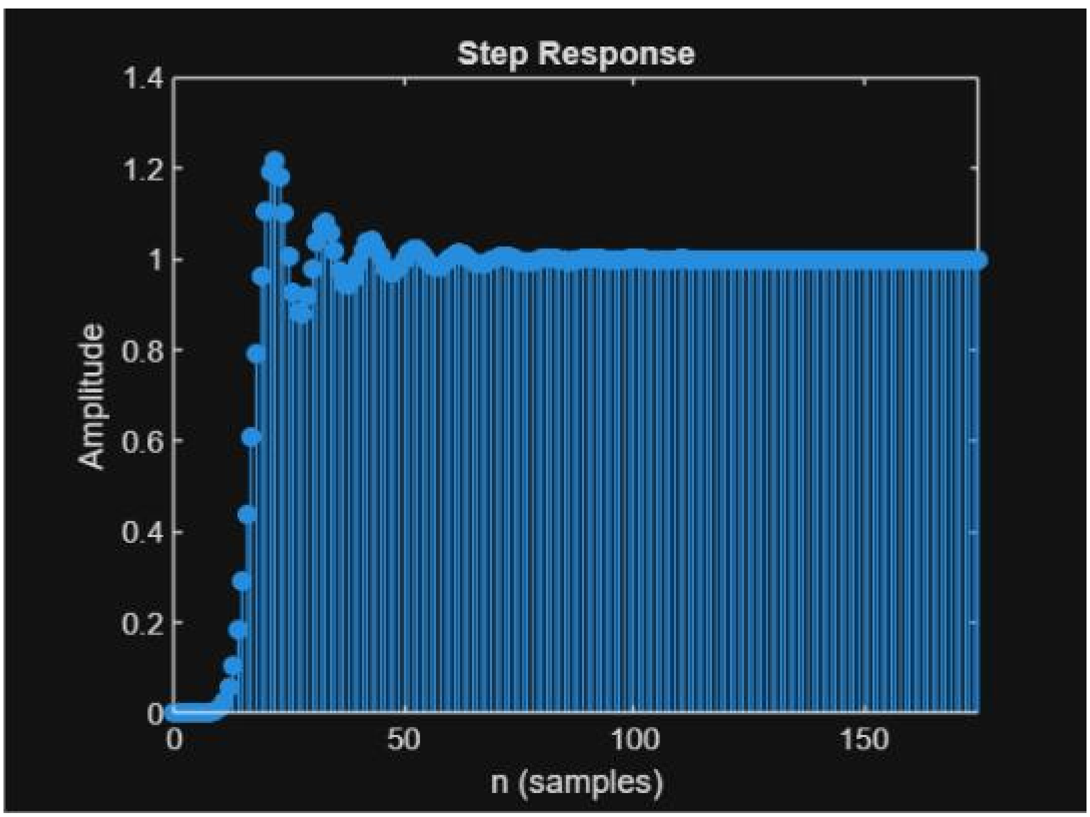
1.1. Design and Analysis of a Butterworth IIR Low-Pass Filter Using MATLAB.

```
clc;
clear;
close all;
Fs = 10000;
Fp = 1000;
Fst = 1500;
[N,Wn] = buttord(Fp/(Fs/2),Fst/(Fs/2),1,60);
[b,a] = butter(N,Wn);
% Frequency Response
figure;
freqz(b,a);
% Impulse Response
[h,n] = impz(b,a,50);
figure;
stem(n,h,'filled');
grid on;
xlabel('Samples');
ylabel('Amplitude');
title('Impulse Response');
```

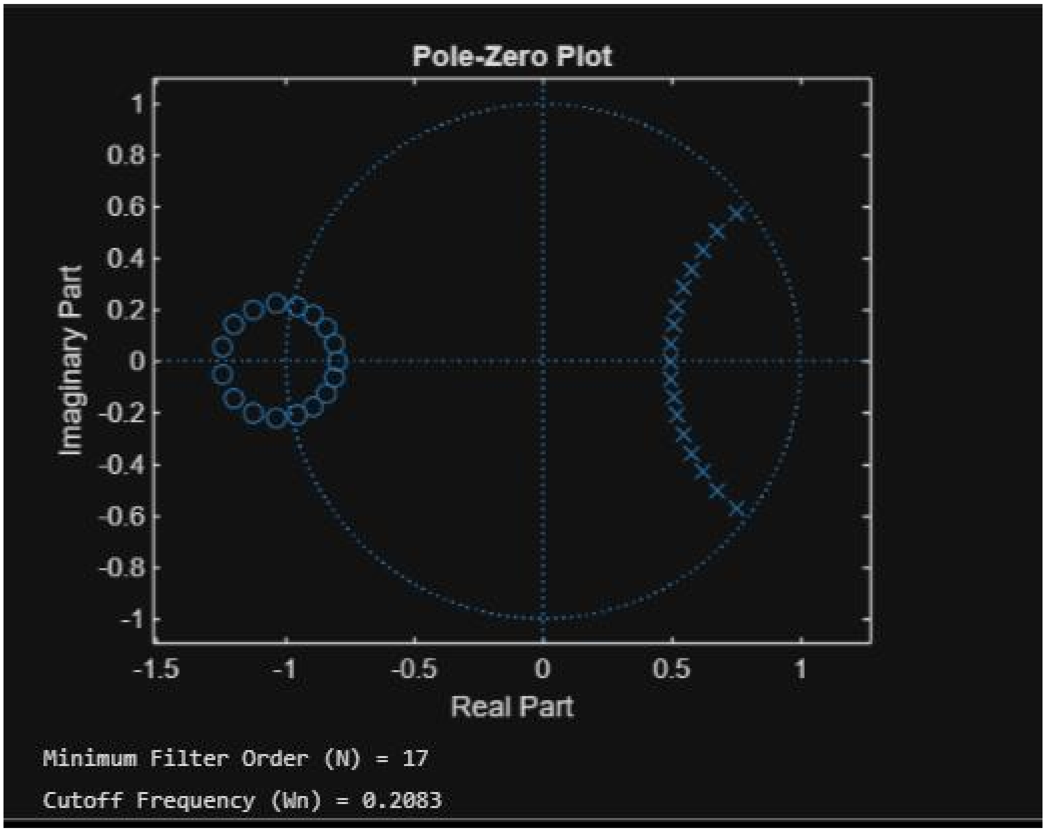




Impulse Response
Amplitude
Samples
0.2
0.15
0.1
0.05
0
-0.05
-0.1
0
10
20
30
40



Step Response
Amplitude
n (samples)
1.4
1.2
1
0.8
0.6
0.4
0.2
0
0
50
100
150



Result:

1.The minimum filter order and cutoff frequency satisfying the given specifications were determined successfully.

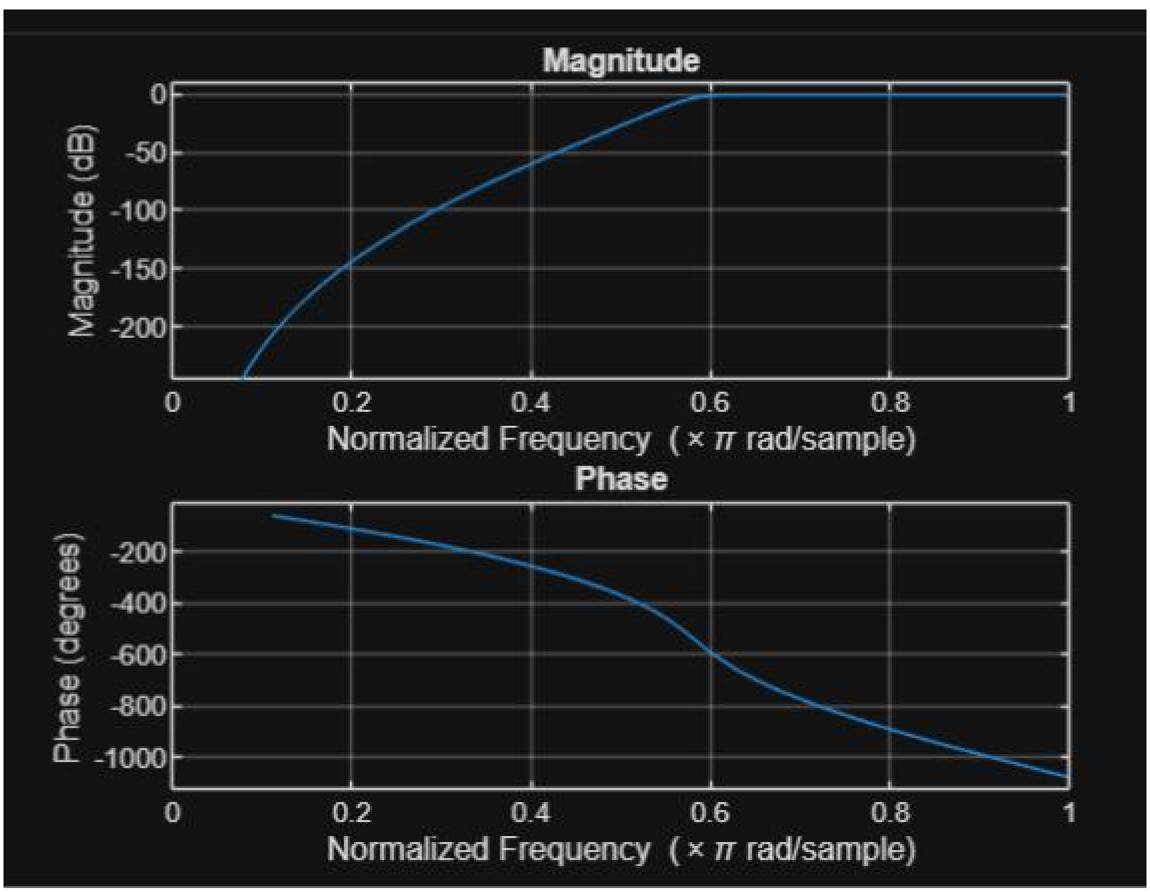
2.The magnitude and phase responses of the Butterworth IIR low-pass filter were obtained and analyzed.

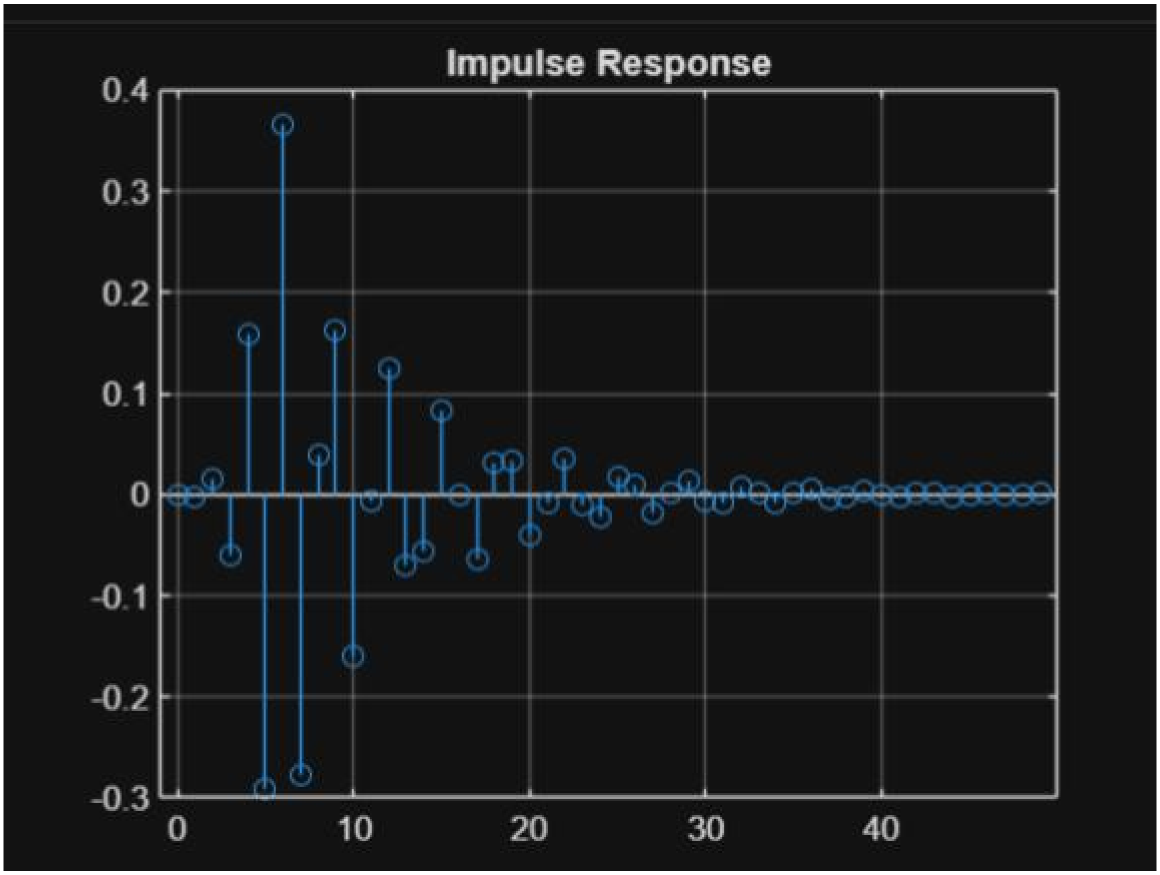
3.The impulse response, step response, and group delay characteristics were evaluated successfully.

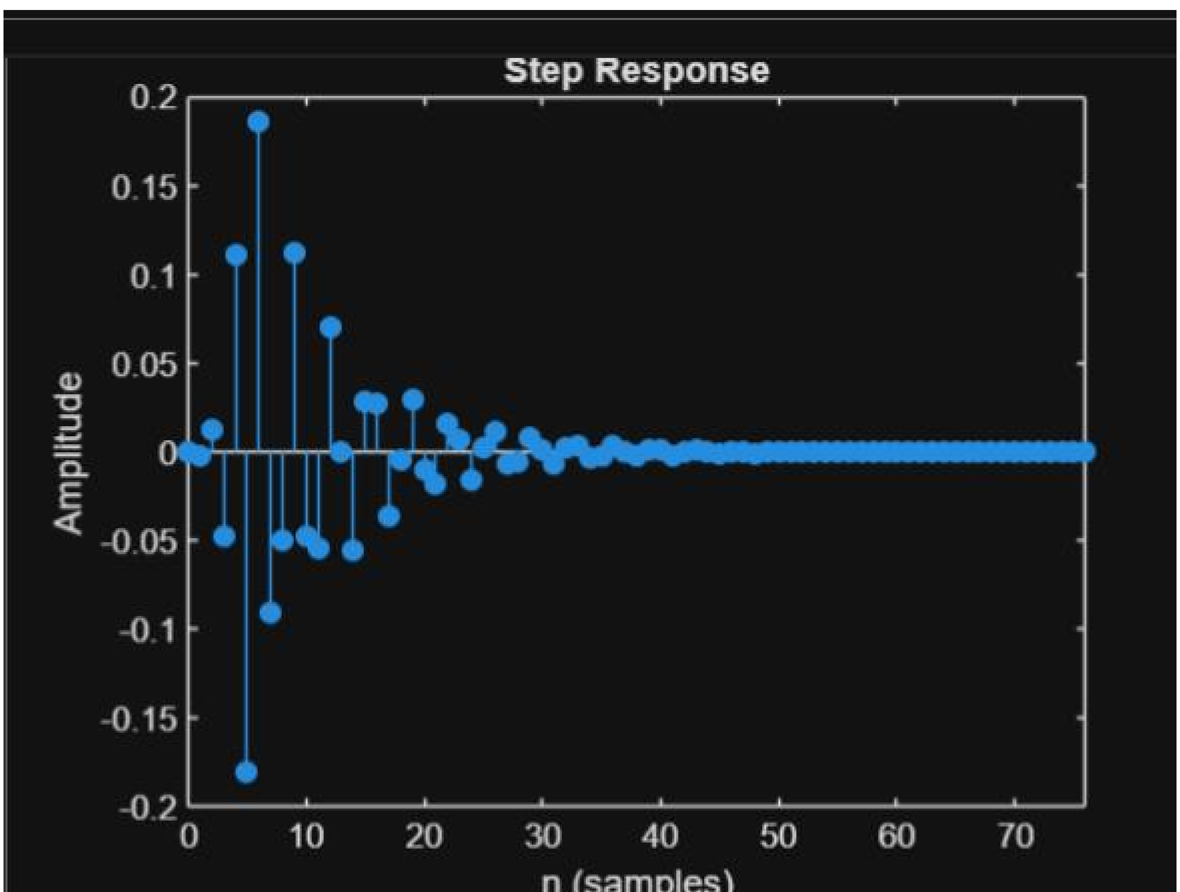
4. The pole-zero plot confirmed the stability and proper performance of the designed filter.

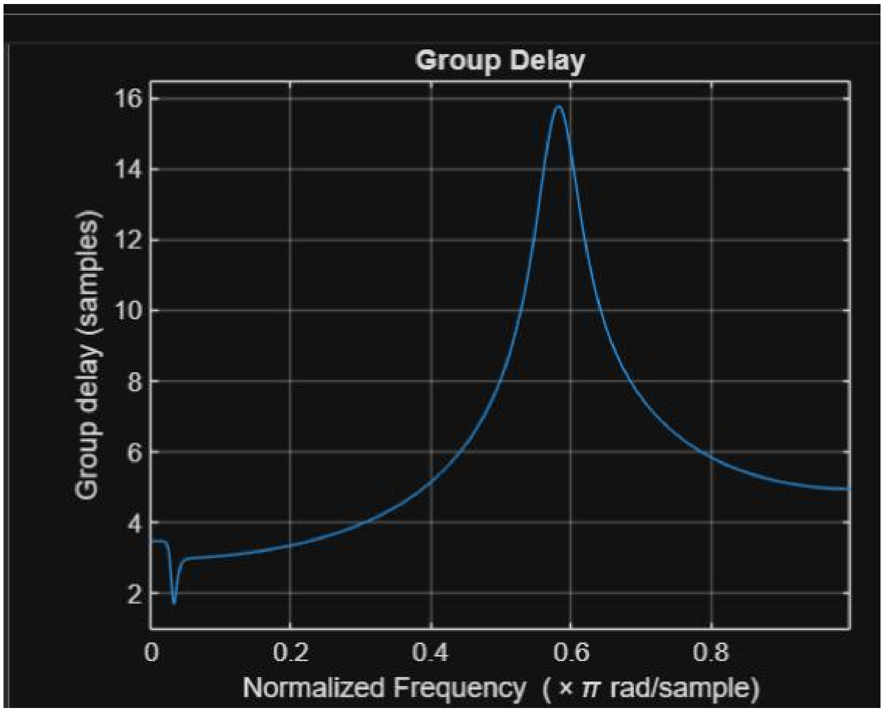
1.2. Design and Analysis of a Butterworth IIR High-Pass Filter Using MATLAB.

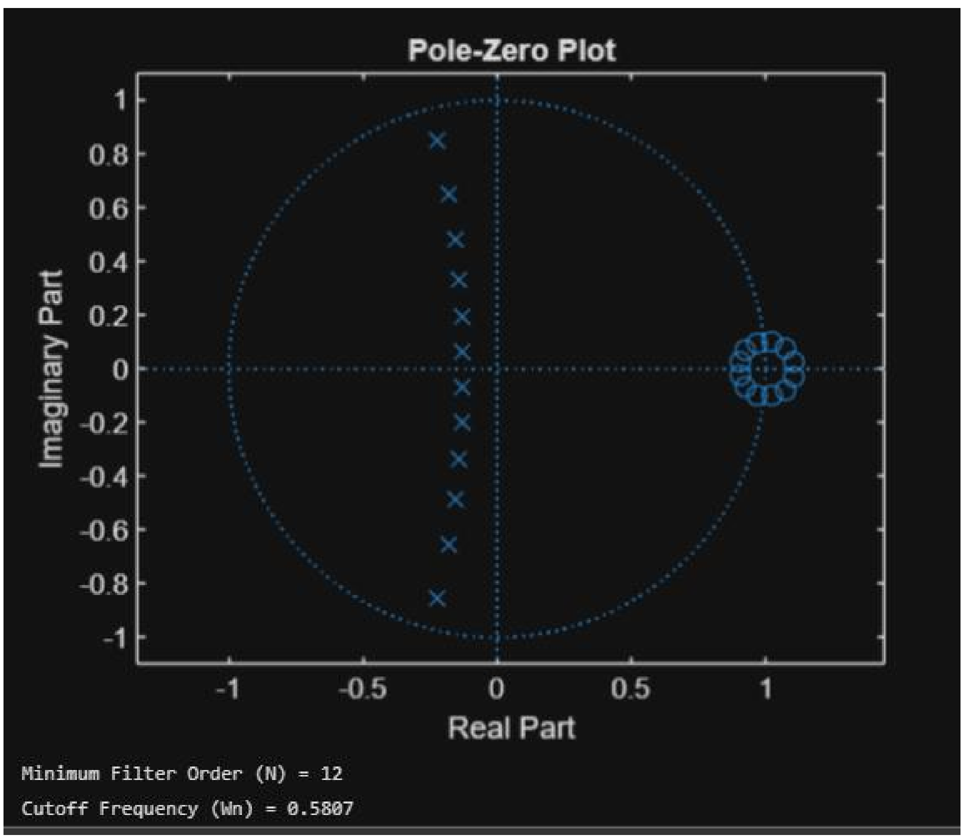
```
clc; clear
Fs=10000; Fp=3000; Fst=2000;
[N,Wn]=buttord(Fp/(Fs/2),Fst/(Fs/2),1,60);
[b,a]=butter(N,Wn,'high');
% Magnitude & Phase
freqz(b,a)
% Impulse Response
figure
[h,n]=impz(b,a,50);
stem(n,h), grid on
title('Impulse Response')
% Step Response
figure
stepz(b,a)
title('Step Response')
% Group Delay
figure
grpdelay(b,a)
title('Group Delay')
% Pole-Zero Plot
figure
zplane(b,a)
title('Pole-Zero Plot')
fprintf('Minimum Filter Order (N) = %d\n',N);
fprintf('Cutoff Frequency (Wn) = %.4f\n',Wn);
```




Impulse Response
0.4
0.3
0.2
0.1
0
-0.1
-0.2
-0.3
0
10
20
30
40


Step Response
Amplitude
0.2
0.15
0.1
0.05
0
-0.05
-0.1
-0.15
-0.2
0
10
20
30
40
50
60
70
n (samples)


Group Delay
Group delay (samples)
16
14
12
10
8
6
4
2
0
0.2
0.4
0.6
0.8
Normalized Frequency (× π rad/sample)


Pole-Zero Plot
Imaginary Part
1
0.8
0.6
0.4
0.2
0
-0.2
-0.4
-0.6
-0.8
-1
-0.5
0.5
Real Part
Minimum Filter Order (N) = 12
Cutoff Frequency (Wn) = 0.5807

Result:

1.The minimum filter order and cutoff frequency satisfying the given specifications were determined successfully.

2.The magnitude and phase responses of the Butterworth IIR High-pass filter were obtained and analyzed.

3.The impulse response, step response, and group delay characteristics were evaluated successfully.

4. The pole-zero plot confirmed the stability and proper performance of the designed filter.